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Roll No

EX-5001-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Electro Magnetic Field Theory

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any missing data.

1. a) Describe Cartesian coordinate system. How will you convert Cartesian system into cylindrical coordinate system? Explain.
b) Explain gradient, divergence and curl of a vector field.
2. a) Describe Coulomb's law. Determine the electric field due to line charge.
b) Derive Poisson's and Laplace's equations and discuss their significance.
3. a) In free space $V = 6xy^2z + 8$, at point $P(1, 2, -5)$, find $\vec{E}$ and P_v .
b) State and verify the boundary value conditions for electric field.
4. a) Determine the magnetic field intensity $\vec{H}$ at the centre of a square current element. The length of each side is 2m and the current $I = 1.0\text{Amp}$.
b) Calculate the magnetic flux density produced by a current loop of radius R on the loop axis when the loop is carrying a current I and situated in air.

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5.
 - a) Derive an expression for transmission coefficient of a uniform plane wave propagation of uniform plane waves with normal incidence at plane dielectric boundary.
 - b) What do you mean by total internal reflection explain?
6.
 - a) Derive an expression for transmission line terminated by a load impedance Z_L .
 - b) Show that $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$.
7.
 - a) Derive expression for force between parallel wires carrying currents in the same direction.
 - b) Two long parallel wires separated 7m apart carry currents of 55A and 105A respectively in the same direction. Determine the magnitude of the force between them per unit length.
8. Write the short note on (any three):
 - i) Transmission line equation
 - ii) Propagation constant of transmission line
 - iii) Standing wave ratio
 - iv) Distortion less and Loss less line.
